

SIMATS ENGINEERING

ASSESSMENT TOOL 1

NAME: N.Vijetha

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COURSE NAME: CSA16Data Warehousing and Data Mining COURSE CODE: CSA1603

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Data Interpretation

Criteria	Understanding of Data (2.5)	Analysis (2.5)	Application of Concepts (2)	Conclusion (1.5)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I N. Vijetha / 192311258 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N.Vijetha

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID Age Monthly Medical Expense (₹) Health Risk Score

P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K SSE

1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and

determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Dataset: Hospital patient segmentation

patient ID	Age	Monthly medical Expense (£)	Health Risk Score
P ₁	28	12000	65
P ₂	55	25000	40
P ₃	42	20000	75
P ₄	30	11000	70
P ₅	60	28000	35

SSE values:

k	SSE
1	600
2	350
3	220
4	200
5	190

Question:

A health-care organisation uses k-means clustering to group patients based on their medical expenses & health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting the number of clusters affects patient segmentation & healthcare decision-making.

SSE - means $\rightarrow$ sum of squared errors

- we are using Elbow method
- The reduction in SSE in

1	600	-
2	350	250
3	220	130
4	200	20
5	190	10

we observe that when
 $k=2$ & 3 the ssf value is been reduced much
& when k value is 4 & 5 the decrease is very small.
This variation happens after the value of k is 3 .
 $\therefore$ Elbow method suggest the value of k is 3 .

$\therefore$ Based on health risk score 3 clusters have formed.

(i) Low risk - with low expense

(ii) Moderate risk - with moderate Expense

(iii) High risk - with high Expense.

Impact on Decision making:

$\Rightarrow$ for low risk group:

(i) preventive care

(ii) Routine monitoring

$\Rightarrow$ for moderate

(i) Regular follow-up

(ii) Targeted Intervention

$\Rightarrow$ for High

(i) Immedation Intervention

(ii) Intensive monitoring

OUT PUT :

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split % 66
- ☐ Classes to clusters evaluation (Num) HealthRisk
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 11:41:37 - SimpleKMeans
- 11:44:04 - SimpleKMeans**

Clusterer output

```

=== Run information ===

Scheme:      weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -
Relation:     hospital
Instances:    5
Attributes:   3
              Age
              MedicalExpense
              HealthRisk
Test mode:    evaluate on training data

=== Clustering model (full training set) ===

kMeans
=====

Number of iterations: 3
Within cluster sum of squared errors: 0.047086194312283745

Initial starting points (random):

Cluster 0: 30,11000,70
Cluster 1: 55,25000,40
Cluster 2: 28,12000,65

Missing values globally replaced with mean/mode

```

Status OK Log

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split % 66
- ☐ Classes to clusters evaluation (Num) HealthRisk
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 11:41:37 - SimpleKMeans
- 11:44:04 - SimpleKMeans**

Clusterer output

```

Cluster 0: 30,11000,70
Cluster 1: 55,25000,40
Cluster 2: 28,12000,65

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute      Full Data      Cluster#
              (5.0)      (1.0)      (2.0)      (2.0)
=====
Age            43            42            57.5      29
MedicalExpense 19200         20000         26500     11500
HealthRisk     57            75            37.5      67.5

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      1 ( 20%)
1      2 ( 40%)
2      2 ( 40%)

```

Status Log

Weka Clusterer Visualize: 11:44:04 - SimpleKMeans (hospital)

X: Instance_number (Num) Y: Age (Num)

Colour: Cluster (Nom) Select Instance

Reset Clear Open Save Jitter

Plot: hospital_clustered

Class colour

cluster0 cluster1 cluster2

Q2. Dataset: E-Commerce Order Analysis

Order ID Product Category Order Value (₹)

O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior

Q2. Dataset: E-commerce order Analysis

order ID	product category	order value
01	Electronics	55000
02	clothing	25000
03	Electronics	57000
04	furniture	30000
05	clothing	27000

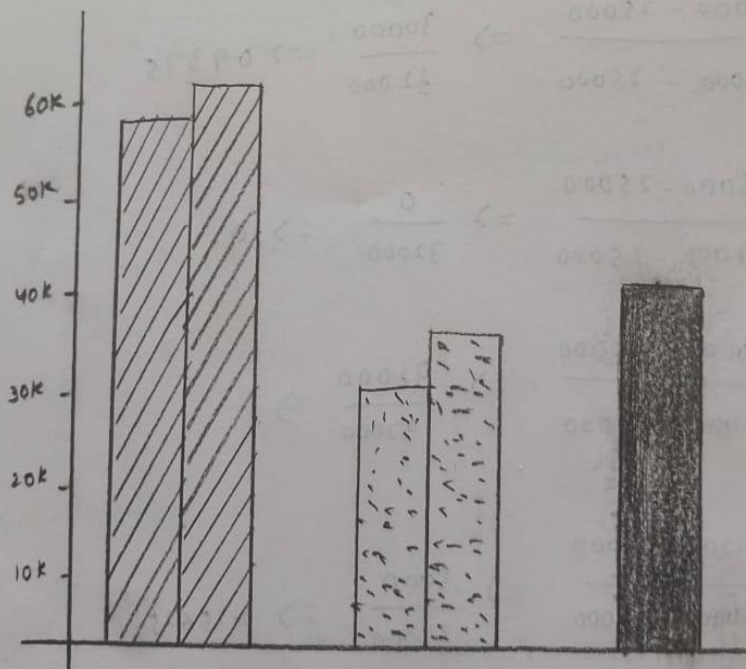
Question:

An online Retailer applies hierarchical clustering to group customer orders based on product category & order value.

A dendrogram is generated from the dataset. Interpret how the clusters are formed & identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

HAC:

- ⇒ It calculates the distance between the clusters
- ⇒ merges the 2 closest clusters
- ⇒ Recalculates the distance
- ⇒ Continue until it will become as one cluster
- ⇒ 01 & 03 forms 1st cluster - Electronics
- ⇒ 02 & 05 forms 2nd cluster - clothing
- ⇒ 04 forms 3rd cluster - furniture



But the optimal number of clusters cannot be determined by the given data.

order	electronics	clothing	
01	1	0	0
02	0	1	0
03	1	0	0
04	0	0	1
05	0	1	0

step-01 : Normalization order value

use min-max Normalization

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Here:

$$X_{\min} = 25,000$$

$$X_{\max} = 57,000$$

$$\therefore O_1 : \frac{55000 - 25000}{57000 - 25000} \Rightarrow \frac{30000}{32000} \Rightarrow 0.9375$$

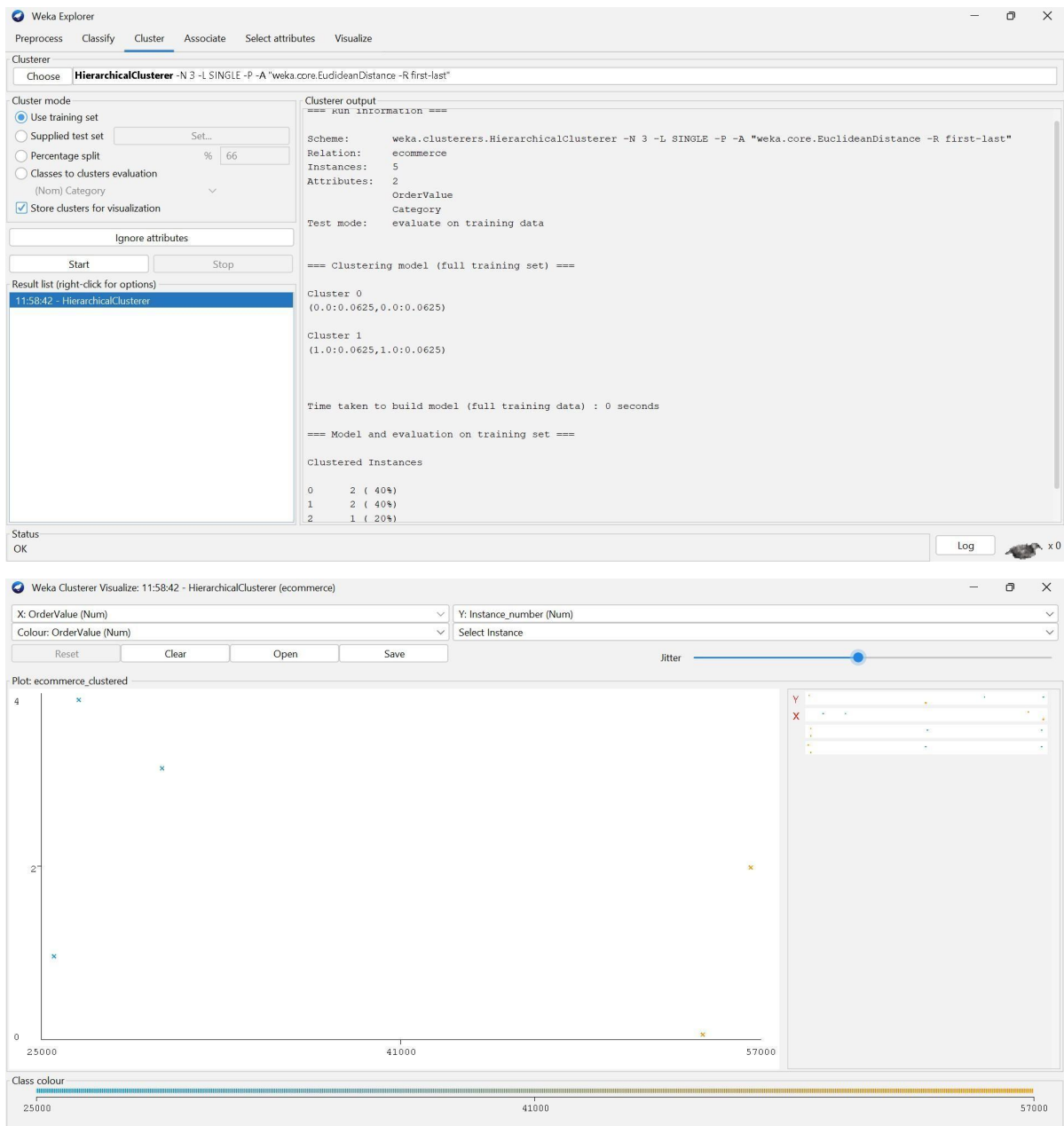
$$O_2 : \frac{25000 - 25000}{57000 - 25000} \Rightarrow \frac{0}{32000} \Rightarrow 0$$

$$O_3 : \frac{57000 - 25000}{57000 - 25000} \Rightarrow \frac{32000}{32000} \Rightarrow 1$$

$$O_4 : \frac{30000 - 25000}{57000 - 25000} \Rightarrow \frac{5000}{32000} \Rightarrow 0.15625$$

$$O_5 : \frac{27000 - 25000}{57000 - 25000} \Rightarrow \frac{2000}{32000} \Rightarrow 0.0625$$

OUTPUT :



Q3. Dataset: Mobile App User Behaviour

User Daily Usage Time (min) Login Frequency In-App Purchase (₹)

U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q₃ Dataset: Mobile APP user Behavior

User	Daily usage time	Login frequency	In-APP purchase
U ₁	120	18	2000
U ₂	60	5	5000
U ₃	140	22	1500
U ₄	50	4	6000
U ₅	130	20	1800

Question:

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Step 1: Identify features:

We have 3 features for each user:

- usage time
- Login frequency
- Purchase Amount

Step 2: Observe the data

Look at the users behavior

U₁, U₃, U₅:

- High usage
- High Login frequency
- Lower Purchase amount

So they form one similar group

U₂, U₄:

- Low usage

- Low Login frequency
- High Purchase amount

so they form another similar group

step-03: compare the two groups

features	$G_1: u_1, u_3, u_5$	$G_2: u_2, u_4$
usage	High	Low
logins	High	Low
purchase	low	High

$\therefore$ There is a clear difference between the two groups.

Step-4: Determine no. of clusters

Since, there are 2 distinct behavioral patterns

we choose: $k = 2$

$\therefore$ no. of clusters = 2

step-5: Name the clusters

cluster 01 - Highly engaged, lower-purchase users

$\rightarrow u_3, u_1, u_5$

cluster 02 - Less engaged, higher-purchase users

$\rightarrow u_2, u_4$

step-6: final results

$k = 2$

cluster = 1 = $\{u_1, u_3, u_5\}$

cluster = 2 = $\{u_2, u_4\}$

I analyzed three features usage, login frequency & purchase amount. The users show two clearly different behavioural patterns. $\therefore$ The appropriate no. of clusters formed is "2".

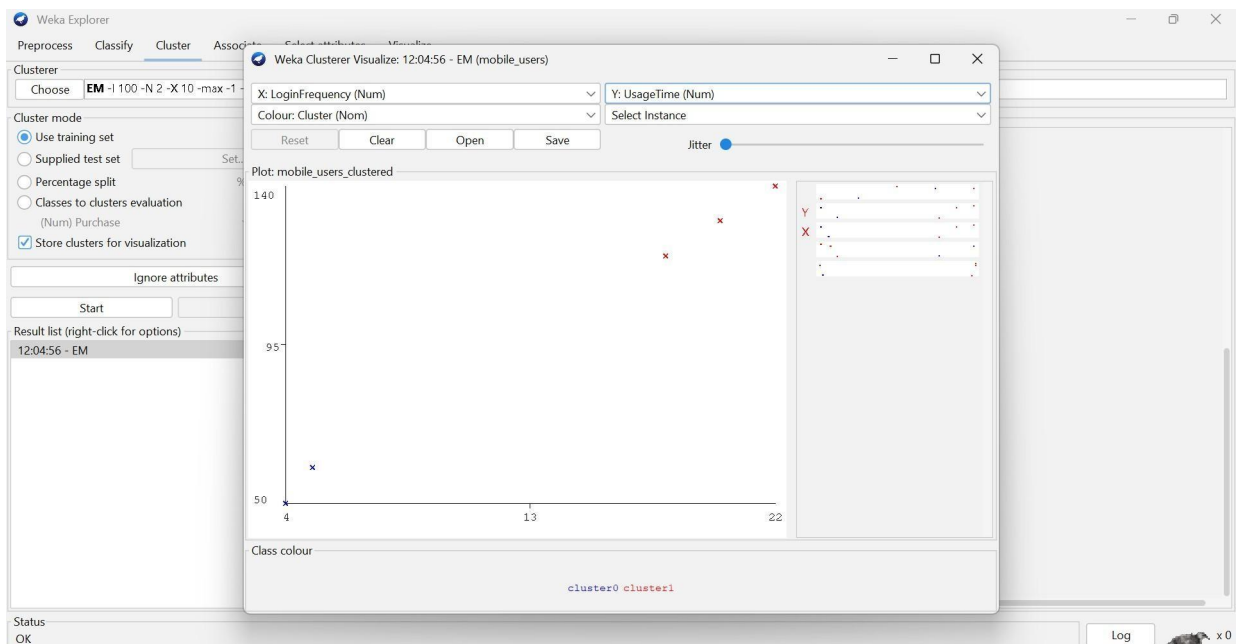
OUTPUT:

The screenshot shows the Weka Explorer interface with the 'Cluster' tab selected. The 'Clusterer' dropdown is set to 'EM'. The 'Cluster mode' section has 'Use training set' selected. The 'Ignore attributes' field is empty. The 'Start' button is visible. The 'Result list' shows '12:04:56 - EM'. The 'Clusterer output' pane displays the following information:

```
=== Run information ===  
  
Scheme:      weka.clusterers.EM -I 100 -N 2 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 10  
Relation:     mobile_users  
Instances:    5  
Attributes:   3  
              UsageTime  
              LoginFrequency  
              Purchase  
Test mode:    evaluate on training data  
  
=== Clustering model (full training set) ===  
  
EM  
===  
  
Number of clusters: 2  
Number of iterations performed: 2  
  
Attribute      Cluster      1  
                (0.4)      (0.6)  
-----  
UsageTime  
mean           55      130  
std. dev.      5      8.165  
  
LoginFrequency  
mean           4.5      20  
std. dev.      0.5      1.633  
  
Purchase  
mean           5500 1766.6667  
std. dev.      500 205.4805
```

The screenshot shows the Weka Explorer interface with the 'Cluster' tab selected. The 'Clusterer' dropdown is set to 'EM'. The 'Cluster mode' section has 'Use training set' selected. The 'Ignore attributes' field is empty. The 'Start' button is visible. The 'Result list' shows '12:04:56 - EM'. The 'Clusterer output' pane displays the following information:

```
=====  
UsageTime  
mean           55      130  
std. dev.      5      8.165  
  
LoginFrequency  
mean           4.5      20  
std. dev.      0.5      1.633  
  
Purchase  
mean           5500 1766.6667  
std. dev.      500 205.4805  
  
Time taken to build model (full training data) : 0.01 seconds  
  
=== Model and evaluation on training set ===  
  
Clustered Instances  
  
0      2 ( 40%)  
1      3 ( 60%)  
  
Log likelihood: -12.53156
```



Q4. Dataset: Agricultural Crop Monitoring

Farm Rainfall (mm) Soil Fertility Score Crop Yield (tons)

F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how Normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q4. Dataset: Agricultural crop monitoring

Farm	Rainfall (mm)	Soil fertility score	Crop yield (tons)
F ₁	300	8	85
F ₂	500	4	70
F ₃	400	6	80
F ₄	280	9	90
F ₅	520	3	65

Question:

An agricultural research center performs clustering on farm data before & after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation & improves the reliability of agricultural data analysis

Step 1: calculate distances before normalization

we use Euclidean distance

$$d(P, Q) = \sqrt{\sum_{i=1}^n (P_i - Q_i)^2}$$

$$d(F_1, F_4) = \sqrt{(300-280)^2 + (8-9)^2 + (85-90)^2}$$

$$= \sqrt{400 + 1 + 25}$$

$$= \sqrt{426} \approx 20.64$$

The important distances are

Pair	Distance
F ₁ - F ₄	20.64
F ₂ - F ₅	20.64
F ₁ - F ₃	100.14

$F_2 - F_3$	100.52
$F_3 - F_4$	120.45
$F_3 - F_5$	120.97
$F_1 - F_2$	200.60
$F_4 - F_5$	241.37

HAC before normalization

using single Linkage, the smallest distance is:

$$F_1 - F_4 = 20.64$$

$$\Delta F_2 - F_5 = 20.64$$

$$\therefore C_1 = \{F_1, F_4\}$$

$$C_2 = \{F_2, F_5\}$$

$$C_3 = \{F_3\}$$

Next, F_3 joins

C_1 , because:

$$\min[d(F_3, F_1), d(F_3, F_4)] = \min(100.14, 120.45) \\ = 100.14$$

So:

$$C_1 = \{F_1, F_3, F_4\}$$

Finally, F_2 connects to C_2 at 100.52, producing one cluster.

Step 2: min-max normalization:

The min-max formula is:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

• Rainfall: $x_{\min} = 280$, $x_{\max} = 520$

• Soil fertility: $x_{\min} = 3$, $x_{\max} = 9$

• crop yield: $x_{\min} = 65$, $x_{\max} = 90$

farm	Rainfall	soil fertility	crop yield
F_1	0.0833	0.8333	0.8000
F_2	0.9167	0.1667	0.2000
F_3	0.5000	0.5000	0.6000
F_4	0.0000	1.0000	1.0000
F_5	1.0000	0.0000	0.0000

step-3: Euclidean distance

pair	Distance
$F_1 - F_4$	0.2734
$F_2 - F_5$	0.2734
$F_1 - F_3$	0.5698
$F_2 - F_3$	0.6669
$F_3 - F_4$	0.8124
$F_3 - F_5$	0.9274

$\therefore$ smallest distance are $F_1 - F_4$ & $F_2 - F_5$

step-4: HAC clustering

At 3 clusters:

$$C_1 = \{F_1, F_4\}$$

$$C_2 = \{F_2, F_5\}$$

$C_3 = \{F_3\}$

$\therefore$ similar forms are grouped together

Step 5: Interpretation

Before normalization, rainfall dominates all because its numerical range is much larger.

After normalization, all features are scaled to 0-1 giving rainfall, soil fertility & crop yield to more balanced influence

$\therefore$ The final 3 clusters:

$\{F_1, F_4\}, \{F_2, F_5\}, \{F_3\}$

OUTPUT:

The top screenshot shows the Weka Explorer interface with the SimpleKMeans algorithm selected. The 'Cluster mode' is set to 'Use training set'. The 'Result list' shows '12:15:11 - SimpleKMeans'. The 'Clusterer output' pane displays the following information:

```

=== Run information ===

Scheme:      weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Relation:    agriculture
Instances:   5
Attributes:  3
              Rainfall
              SoilFertility
              CropYield
Test mode:   evaluate on training data

=== Clustering model (full training set) ===

kMeans
=====

Number of iterations: 3
Within cluster sum of squared errors: 0.07472222222222222

Initial starting points (random):

Cluster 0: 280,9,90
Cluster 1: 500,4,70
Cluster 2: 300,8,85
  
```

The bottom screenshot shows the same Weka Explorer interface, but the 'Clusterer output' pane displays the following information:

```

Cluster 0: 280,9,90
Cluster 1: 500,4,70
Cluster 2: 300,8,85

Missing values globally replaced with mean/mode

Final cluster centroids:

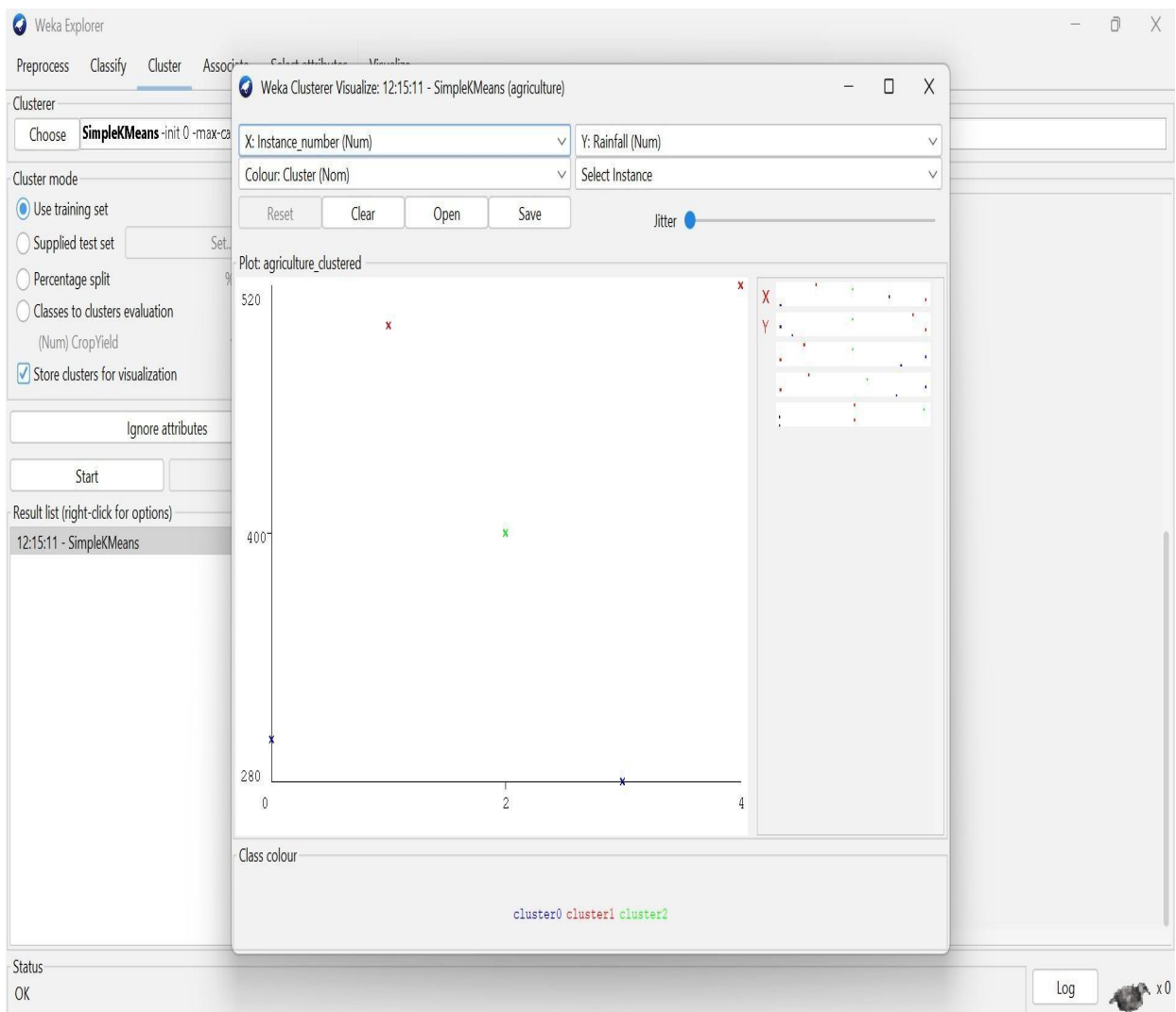
Attribute      Full Data      Cluster#
              (5.0)         0         1         2
              (2.0)         (2.0)        (1.0)
-----
Rainfall       400           290           510           400
SoilFertility   6             8.5           3.5           6
CropYield       78            87.5          67.5          80

Time taken to build model (full training data) : 0.02 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      2 ( 40%)
1      2 ( 40%)
2      1 ( 20%)
  
```



Q5. Dataset: Employee Performance Evaluation

Employee Productivity Score Attendance (%) Skill Assessment Score

E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model Score

K=2	0.48
K=3	0.68
K=4	0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

Q5 Dataset: Employee performance Evaluation

Employee	productivity score	Attendance	skill Assessment score
E ₁	88	92	84
E ₂	62	95	68
E ₃	78	88	72
E ₄	55	65	58
E ₅	93	96	90

silhouette scores

Model score

k=2 0.48

k=3 0.68

k=4 0.52

Question:

A company clusters employees based on productivity, attendance & skill assessment scores. Analyze the silhouette scores obtained for different clustering models &

determine the most suitable no. of clusters justify your answer & Explain how the selected model contributes to effective employee performance evaluation

Given silhouette scores

Model	score
$K=2$	0.48
$K=3$	0.68
$K=4$	0.52

* Analysis:

The silhouette score measures the quality of clustering. A higher score indicates better-defined & well-separated clusters.

- $K=2$: score = 0.48
↳ moderate clustering.

- $K=3$: score = 0.68
↳ Best clustering

- $K=4$: score = 0.52
↳ moderate clustering

Since $K=3$, has the highest silhouette score (0.68), it is the most suitable model.

Justification:

$K=3$ provides better similarity among employees within the same cluster & better separation between different clusters. It can effectively group employees based on their productivity, attendance & skill assessment.

conclusion:

Best number of clusters = 3

The selected $k=3$ model can help the company identify different employee performance groups (such as high, medium & low performers) This supports better performance evaluation, training, recognition & employee development decisions.

Output:

Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize

Clusterer
Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Skill
☒ Store clusters for visualization

Ignore attributes
Start Stop

Result list (right-click for options)
12:15:11 - SimpleKMeans
12:19:50 - SimpleKMeans

Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -
Relation: employees
Instances: 5
Attributes: 3
Productivity
Attendance
Skill
Test mode: evaluate on training data
=== Clustering model (full training set) ===
kMeans
=====

Number of iterations: 2
Within cluster sum of squared errors: 0.27815523773183715
Initial starting points (random):
Cluster 0: 55,65,58
Cluster 1: 62,75,68
Cluster 2: 88,92,84
Missing values globally replaced with mean/mode

Status OK Log

Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize

Clusterer
Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Skill
☒ Store clusters for visualization

Ignore attributes
Start Stop

Result list (right-click for options)
12:15:11 - SimpleKMeans
12:19:50 - SimpleKMeans

Weka Clusterer Visualize: 12:19:50 - SimpleKMeans (employees)
X: Instance_number (Num) Y: Productivity (Num)
Colour: Cluster (Nom) Select Instance
Reset Clear Open Save Jitter

Plot: employees_clustered

Class colour
cluster0 cluster1 cluster2

Status OK Log

Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize

Clusterer
Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Skill
☒ Store clusters for visualization

Ignore attributes
Start Stop

Result list (right-click for options)
12:15:11 - SimpleKMeans
12:19:50 - SimpleKMeans

Clusterer output
Cluster 0: 55,65,58
Cluster 1: 62,75,68
Cluster 2: 88,92,84
Missing values globally replaced with mean/mode
Final cluster centroids:
Attribute Full Data Cluster#
(5.0) (1.0) (1.0) (3.0)
=====

Attribute	Full Data	Cluster# 0	Cluster# 1	Cluster# 2
Productivity	75.2	55	62	86.3333
Attendance	83.2	65	75	92
Skill	74.4	58	68	82

Time taken to build model (full training data) : 0.01 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 1 (20%)
1 1 (20%)
2 3 (60%)

Status OK Log

